cloudtorch / develop — Step 3 outline

Composite forward model (fitted PCA background $\oplus$ two clouds) + a *per-pixel* loss, **fully differentiable, no regularisation**. A standalone **per-pixel** building block, kept independent of the inversion strategy so either a PINN or a classical spatially-regularised fit can wrap it. Deliverable of `instructions.md` step 3.

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1. OBJECTIVE

Assemble the step-2 background model and the existing two-cloud transfer into a *single* autograd-differentiable forward model for **one pixel** (parameters $\to$ one spectrum), and pair it with a per-pixel χ^2 loss. No penalty terms and no inversion strategy are baked in here (§5): regularisation is a *separate outer layer* chosen later, so this same core serves either a PINN or a classical spatially-regularised inversion. Step 3's job is to make the physics + loss clean, differentiable, and strategy-agnostic.

2. COMPOSITE FORWARD MODEL

$$I_{\text{out}} = \underbrace{\text{cloud}_2\big(\text{cloud}_1(I_{\text{in}})\big)}_{\texttt{model\_synth\_2clouds\_givenbck}}, \qquad I_{\text{in}} = \boldsymbol{\mu} + \mathbf{c}\,\mathbf{V}.$$

i.e. `cmt.model_synth_2clouds_givenbck(x, p_cloud, background(pca, c))`, chaining the step-2 `background()` into the given-background two-cloud routine. Differentiable w.r.t. *both* the background coefficients $\mathbf{c}$ and the cloud parameters $\mathbf{p}_{\text{cloud}}$. The atomic unit is **one pixel** $\to$ **one spectrum**; the routine is simply *batched* over an arbitrary set of pixels with **no coupling between them**. That per-pixel independence is what keeps the model reusable by any outer inversion (a row is one pixel now, one (x, y, t) sample later).

3. PARAMETERS

Per sample: **6 background** coefficients $\mathbf{c}$ + **10 cloud** ($S, \Delta\tau, v_{\text{los}}, \Delta v, \log a$ per cloud) = **16** free. $\log a$ **is now a real free parameter** (initialised $-4, -5$); a per-quantity *freeze switch* is retained — the `make_param(..., grad=)` idiom already toggles `requires_grad` — so it can be frozen again later without touching the physics. Held as plain differentiable tensors now, but the forward takes them as arguments, so in step 4 they are swapped for a neural field's outputs unchanged.

Caveat (why it was frozen): free $\log a$ previously drove the Voigt profile to NaNs. We keep the existing NaN guard and clamp $\log a$ to a safe range (e.g. $[-6, 0]$); the freeze switch is the fallback if it still misbehaves.

4. PER-PIXEL LOSS

$$\chi_i^2 = \sum_{\lambda} \big(O_{i\lambda} - S_{i\lambda}\big)^2, \qquad \mathcal{L} = \tfrac{1}{N} \textstyle\sum_i \chi_i^2.$$

Returns the *per-pixel vector* (χ_i^2) *and* its scalar mean — the vector feeds a PINN's coordinate sampling *or* is summed with spatial penalties in a classical fit. Optional $1/\sigma^2$ or $1/N_\lambda$ weighting. **No `regu*` penalties here** (§5).

5. STRATEGY-AGNOSTIC: THE OUTER LAYER IS CHOSEN LATER

Step 3 commits to *no* inversion strategy and adds *no* penalties. The identical per-pixel model + loss is wrapped later by *one* of:

- **A Classical** — per-pixel free parameters + explicit spatial (and temporal) regularisation on the parameter maps (reuse the `utils.py` `regu*` penalties).
- **B PINN** — parameters are the outputs of a smooth coordinate network $f(x, y, t)$; smoothness is implicit (steps 4–5).

Both call the step-3 core *unchanged*; only the parameter source and the regulariser differ. An unregularised per-pixel fit is *knowingly degenerate* (the background can absorb near-rest core signal — your earlier concern); whichever outer layer you pick supplies the regularisation that resolves it. Init: clouds at $\mp 20\,\text{km}\,\text{s}^{-1}$ (not critical).

Verified (`t = 13`, 15 080 px, CPU): gradients reach *both* $\mathbf{c}$ and $\mathbf{p}_{\text{cloud}}$ with no NaNs; a 400-step *unregularised* Adam fit drops the mean per-pixel χ^2 from 5.18 to 0.11 ($\approx 47\times$, $\sim 1.7\%$ RMS residual) and observed-vs-fitted images agree across the line — **the model, per-pixel loss, and fit routine work**. Two honest notes: the low χ^2 is *not* proof of a correct background/cloud split (unregularised $\Rightarrow$ still degenerate); and free $\log a$ is stable but *weakly constrained* (grad $\lesssim 10^{-5}$), so it barely leaves its init — freezing it would cost almost nothing.

Built & run: `composite_model.py` (`composite_synth` + `chi2_per_pixel`) and the step-3 notebook — compose, per-pixel loss, grad check, smoke-fit, obs-vs-fit image panels. Next: the outer layer (a PINN, steps 4–5, *or* a classical spatially-regularised fit) wraps this unchanged.